Advantages of Longitudinal Studies

- Economizes on subjects
- Subjects serve as own control
- Between-subject variation excluded from error
- Can provide more efficient estimators than cross-sectional designs with same number and pattern of observations
- Can separate aging effects (changes over time within individuals) from cohort effects (differences between subjects at baseline)
  $\Rightarrow$ cross-sectional design can't do this
- Can provide information about individual change

Challenges of Longitudinal Data Analysis

- Observations are not, by definition, independent $\Rightarrow$ must account for dependency in data
- Analysis methods not as well developed, especially for more sophisticated models
- Lack and difficulty of using software
- Computationally intensive
- Unbalanced designs, missing data, attrition
- Time-varying covariates
- Carry-over effects (when repeated factor is condition or treatment, not time)

Notation

- $i = 1, \ldots, N$ subjects
- $j = 1, \ldots, n_i$ observations (n for balanced designs)
- total number of observations $= \Sigma_i^N n_i$
- $\boldsymbol{y}_i = n_i \times 1$ vector of responses
- $\boldsymbol{x}_{ij} = p \times 1$ covariate vector for subject i at time j
  - time-invariant or time-independent covariates (between-subjects)
  - time-varying or time-dependent covariates (within-subjects)
- $\boldsymbol{X}_i = n_i \times p$ matrix of covariates for subject i
  usually includes an intercept term

Data Layout

subject	observation	response	covariates		
1	1	y_{11}	x_{111}	$\ldots$	x_{11p}
1	2	y_{12}	x_{121}	$\ldots$	x_{12p}
.	.	.	.	. .	
1	n_1	y_{1n_1}	x_{1n_11}	$\ldots$	x_{1n_1p}
.	.	.	.	. .	
.	.	.	.	. .	
.	.	.	.	. .	
.	.	.	.	. .	
N	1	y_{N1}	x_{N11}	$\ldots$	x_{N1p}
N	2	y_{N2}	x_{N21}	$\ldots$	x_{N2p}
.	.	.	.	. .	
N	n_N	y_{Nn_N}	x_{Nn_N1}	$\ldots$	x_{Nn_Np}

- n_i varies by subjects (some analyses won't allow this)
- above is "univariate layout"
- different layout for repeated measures MANOVA ("multivariate layout")
- if x_r is time-invariant (between-subjects) $x_{i1r} = x_{i2r} = x_{i3r} = \ldots = x_{in_ir}$

Analysis Considerations

- Response variable
  - continuous (normal or non-normal)
  - categorical (dichotomous, ordinal, nominal, counts)
- Number of subjects N
- Number of observations per subject n_i
  - $n_i = 2$ for all: change score analysis or ANCOVA
  - $n_i = n$ for all: balanced design - ANOVA or MANOVA for repeated measures
  - n_i varies: more general methods

- Number & type of covariates - $E(\boldsymbol{y}_i)$
  - one sample
  - multiple samples
  - regression (continuous or categorical covariates)
  - time-varying covariates
- Type of variance-covariance structure - $V(\boldsymbol{y}_i)$
  - homogeneous or heterogeneous variances
  - homogeneous or heterogeneous covariances

General Approaches

- Derived variable: not really longitudinal, per se, reduce the repeated observations into a summary variable
  - average across time
  - change score
  - linear trend across time
  - last observation
- Longitudinal Analysis
  - ANOVA for repeated measures
  - MANOVA for repeated measures
  - Mixed-effects regression models
  - Covariance pattern models
  - Generalized Estimating Equations (GEE) models

Simplest Longitudinal Analysis

Paired t-test can be used to address whether there is significant average change between two timepoints

- $i = 1, \ldots, N$ subjects
- y_{i1} = pre-test
- y_{i2} = post-test
- $d_i = y_{i2} - y_{i1}$ = post to pre change score

$H_0 : \mu_{y_1} = \mu_{y_2}$ same as $H_0 : (\mu_{y_2} - \mu_{y_1}) = 0$

test statistic

$$
\begin{aligned}
t &= \bar{d} \,/\, \left(s_d/\sqrt{N}\right) \\
&= \bar{d} \,/\, \left(\sqrt{\left[\sum_i d_i^2 - (\sum_i d_i)^2/N\right]/(N-1)} \,/\, \sqrt{N}\right) \\
&\overset{H_0}{\sim} t_{N-1}
\end{aligned}
$$

Notice, can do the same test using regression model

$$d_i = \beta_0 + e_i$$

and testing $H_0 : \beta_0 = 0$

Change Score analysis

Suppose there is a grouping variable

- $x_i = 0$ for controls
- $x_i = 1$ for treatment group

$$d_i = \beta_0 + \beta_1 x_i + e_i$$

- testing $H_0 : \beta_0 = 0$ tests whether the average change is equal to zero for the control group
- testing $H_0 : \beta_1 = 0$ tests whether the average change is equal for the two groups

notice

$$d_i = \beta_0 + \beta_1 x_i + e_i$$

$$y_{i2} - y_{i1} = \beta_0 + \beta_1 x_i + e_i$$

$$y_{i2} = y_{i1} + \beta_0 + \beta_1 x_i + e_i$$

$\Rightarrow$ change score analysis assumes that the slope for $y_{i1} = 1$

Analysis of covariance of post-test scores

$$y_{i2} = \beta_0 + \beta_1 x_i + \beta_2 y_{i1} + e_i$$

- testing $H_0 : \beta_0 = 0$ tests whether the average post-test is equal to zero for the control group subjects with zero pre-test
- testing $H_0 : \beta_1 = 0$ tests whether the post-test is equal for the two groups, given the same value on the pre-test (*i.e.*, conditional on pre-test)
- testing $H_0 : \beta_2 = 0$ tests whether the post-test is related to the pre-test, conditional on group

Change score analysis and ANCOVA answer different questions

- change score: is average change the same between the groups
- ancova: is post-test average the same between groups for sub-populations with the same pre-test values (*i.e.*, is the conditional average the same between the groups)

Which to use?

- depends on the question of interest
- often yield similar conclusions for group effect
- if subjects randomized to group, then ANCOVA is more efficient (*i.e.*, more powerful)
- must be careful in non-randomized settings, where groups are not necessarily similar in terms of pre-test scores
  - Lord's paradox (Bock, 1975; Allison, 1990)

Lord's example

- Sex differences in the "Freshman 15"
- Analysis of difference scores for males and females shows no difference
- Analysis of covariance shows a significant difference
- So, what is going on?

Lord, F. M. (1967). A paradox in the interpretation of group comparisons. *Psychological Bulletin*, 68 (5). 304-305.

-Distributions on top = initial weight distributions
-Distributions on side = final weight distributions
-Ovals = scatter plots of male and female weights

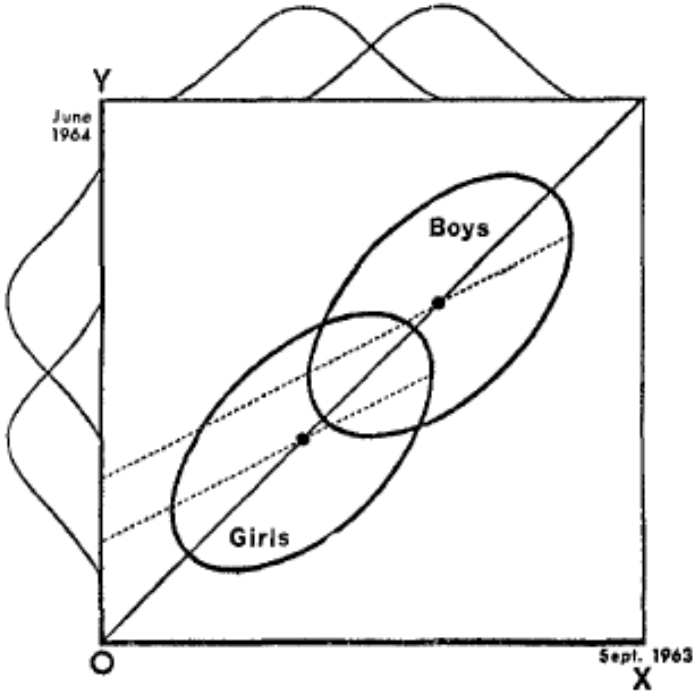


FIG. 1. Hypothetical scatterplots showing initial and final weight for boys and for girls.

Different questions = Different answers

- ANCOVA “asks” how can one variable predict variation in another (in our example how can weight at the beginning be used to predict weight at the end)
- Analysis of difference scores “asks” how do initial scores differ from final scores

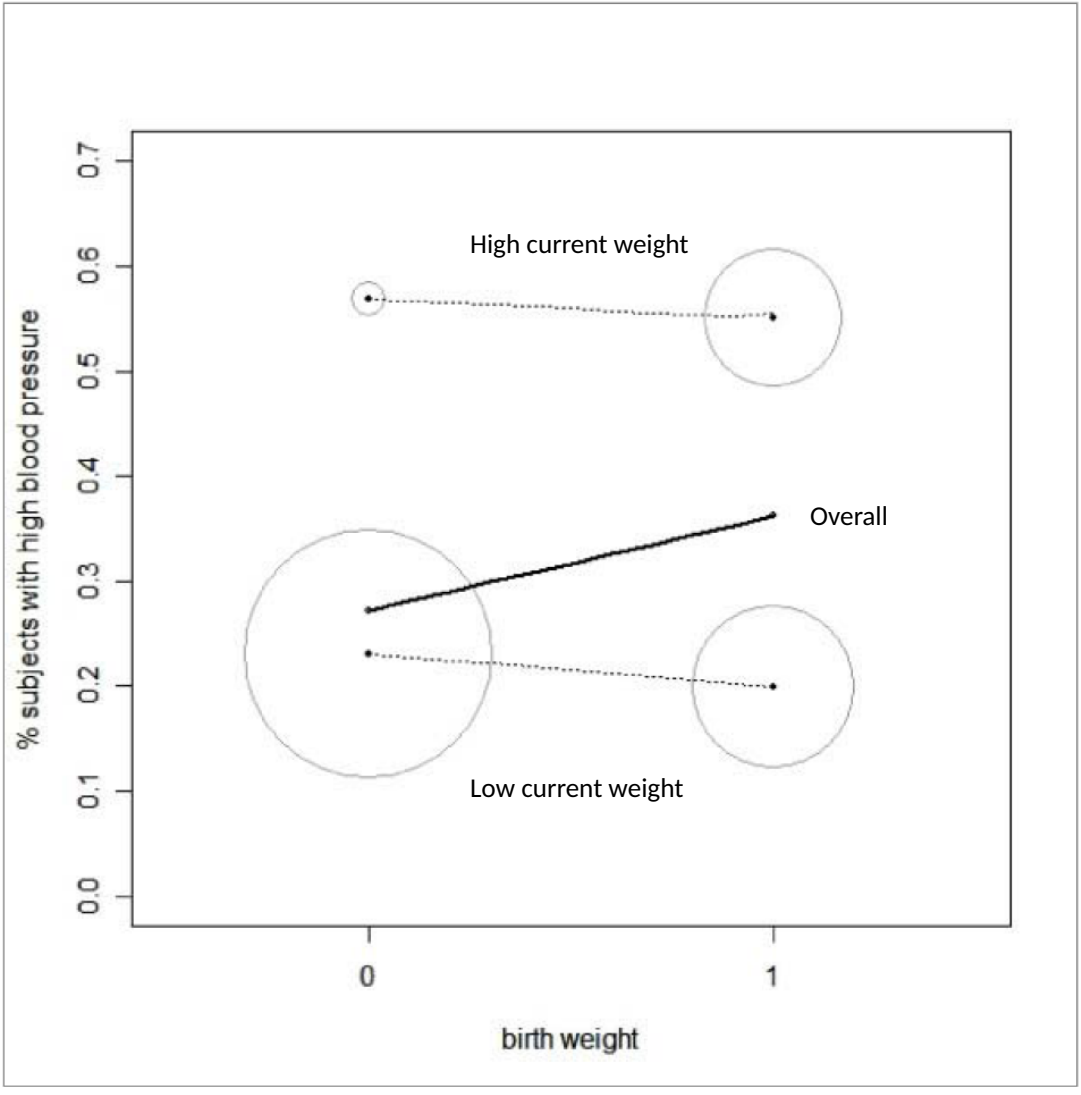
Take Home Message

- When evaluating differences within individuals, there may not be a consistently best way
- Think of the question that you want to ask
- Think of what question the statistical tools can ask
- Be careful in interpreting your results

Simpson's Paradox

Numbers and Percentages of subjects with high blood pressure (> 135 mmHg) according to their birth weight and current body weight

	Normal BP	High BP	Total	Percentage of subjects with high *BP*
Overall:				
Low birth weight	354	132	486	27.2%
High birth weight	328	186	514	36.2%
Total	682	318	1,000	31.8%
Current weight < = 90 Kg				
Low birth weight	329	99	428	23.1%
High birth weight	221	55	276	19.9%
Total	550	154	704	21.9%
Current weight > 90 Kg				
Low birth weight	25	33	58	56.9%
High birth weight	107	131	238	55.0%
Total	132	164	296	55.4%



SuperMix Simpson example

birth_weight	current_weight	interaction	high_bp	n
0	0	0	99	428
1	0	0	55	276
0	1	0	33	58
1	1	1	131	238

Log likelihood = **-59.025343**

Wald chi2(**1**) = **9.34**
Prob > chi2 = **0.0022**

high_bp	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
birth_weight	**1.520787**	**.2086625**	**3.06**	**0.002**	**1.162191**	**1.990028**
_cons	**.3728814**	**.0380277**	**-9.67**	**0.000**	**.3053249**	**.4553854**

Log likelihood = **-11.160946**

Wald chi2(**2**) = **101.60**
Prob > chi2 = **0.0000**

high_bp	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
birth_weight	**.8549828**	**.136181**	**-0.98**	**0.325**	**.6257179**	**1.168251**
current_weight	**4.742032**	**.7768026**	**9.50**	**0.000**	**3.439754**	**6.537349**
_cons	**.297248**	**.0323062**	**-11.16**	**0.000**	**.2402186**	**.3678165**

Log likelihood = **-11.107654**

Wald chi2(**3**) = **101.65**
Prob > chi2 = **0.0000**

high_bp	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
birth_weight	**.8270488**	**.1565882**	**-1.00**	**0.316**	**.5706503**	**1.19865**
current_weight	**4.386667**	**1.267159**	**5.12**	**0.000**	**2.490299**	**7.727123**
interaction	**1.121457**	**.3935173**	**0.33**	**0.744**	**.5637639**	**2.230836**
_cons	**.3009119**	**.0344942**	**-10.48**	**0.000**	**.2403611**	**.3767162**

Berkeley Gender Bias Lawsuit
Graduate School Admissions

	Applicants	Admitted
Men	8442	**44%**
Women	4321	35%

Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	**82%**
B	560	63%	25	**68%**
C	325	**37%**	593	34%
D	417	33%	375	**35%**
E	191	**28%**	393	24%
F	373	6%	341	**7%**

SuperMix Berkeley Example

Department	Gender	Admit	n
1	0	512	825
1	1	89	108
2	0	353	560
2	1	17	25
3	0	120	325
3	1	202	593
4	0	138	417
4	1	131	375
5	0	54	191
5	1	94	393
6	0	22	373
6	1	24	341

Simple Logistic Regression Model

```
. meglm admit gender, family(binomial n) link(logit) or

Iteration 0:   Log likelihood = -434.62055
Iteration 1:   Log likelihood = -425.55478
Iteration 2:   Log likelihood = -425.55446
Iteration 3:   Log likelihood = -425.55446

Mixed-effects GLM                               Number of obs     =         12
Family: Binomial
Link:   Logit
Binomial variable: n

                                                Wald chi2(1)      =      91.71
Log likelihood = -425.55446                     Prob > chi2       =     0.0000
------------------------------------------------------------------------------
       admit | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      gender |   .5423429   .0346509    -9.58   0.000     .4785086    .6146929
       _cons |   .8036193   .0311683    -5.64   0.000     .7447947    .8670899
------------------------------------------------------------------------------
Note: _cons estimates baseline odds (conditional on zero random effects).

. est store m1
```

Mixed-effects Logistic - Random Intercept Model

```
. meglm admit gender || department: , family(binomial n) link(logit) or
```

Mixed-effects GLM		Number of obs	= 12
Family: Binomial			
Link: Logit			
Binomial variable: n			
Group variable: department		Number of groups	= 6
		Obs per group:	
		min	= 2
		avg	= 2.0
		max	= 2
Integration method: mvaghermite		Integration pts.	= 7
		Wald chi2(1)	= 1.23
Log likelihood = -63.070536		Prob > chi2	= 0.2671

admit	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
gender	1.093689	.0882632	1.11	0.267	.9336846	1.281114
_cons	.5047117	.2297442	-1.50	0.133	.2068128	1.231713
department						
var(_cons)	1.228047	.7183762			.390197	3.864971

Fixed-effects Logistic Regression Model

```
. meglm admit gender i.department, family(binomial n) link(logit) or
```

Mixed-effects GLM — Number of obs = 12
Family: Binomial
Link: Logit
Binomial variable: n

Wald chi2(6) = 622.62
Log likelihood = -44.784897 — Prob > chi2 = 0.0000

admit	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
gender	1.100104	.0888861	1.18	0.238	.9389842	1.28887
department						
2	.9572262	.1051402	-0.40	0.691	.771827	1.18716
3	.2836011	.0302334	-11.82	0.000	.2301257	.3495029
4	.274457	.029039	-12.22	0.000	.2230554	.3377038
5	.1776958	.0223753	-13.72	0.000	.1388336	.2274363
6	.0367082	.0062389	-19.44	0.000	.0263083	.0512193
_cons	1.790607	.1235389	8.44	0.000	1.564133	2.049874

Mixed-effects Logistic - Random Intercept and Gender Model

```
. meglm admit gender || department: gender, family(binomial n) link(logit) covariance(unstructured) or
```

```
Mixed-effects GLM                               Number of obs     =         12
Family: Binomial
Link:   Logit
Binomial variable: n
Group variable: department                      Number of groups  =          6

                                                Obs per group:
                                                              min =          2
                                                              avg =        2.0
                                                              max =          2

Integration method: mvaghermite                 Integration pts.  =          7

                                                Wald chi2(1)      =       1.41
Log likelihood = -53.878449                     Prob > chi2       =     0.2354
```

admit	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
gender	1.188898	.1733699	1.19	0.235	.8933443	1.582232
_cons	.485142	.0746623	-4.70	0.000	.3588151	.6559444
department						
var(gender)	.1220386	.099178			.0248168	.6001355
var(_cons)	1.29487	.8131789			.3781565	4.433851
department						
cov(gender,_cons)	.2404408	.1366257	1.76	0.078	-.0273407	.5082223

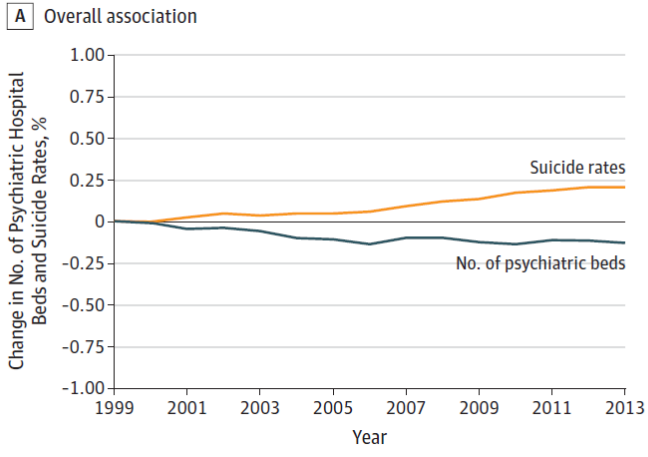
Lord vs. Simpson

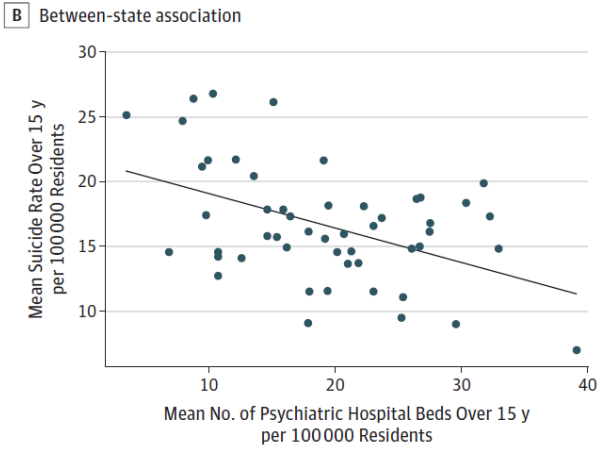
- Lord – binary predictor continuous outcome
  - Difference in conditional and unconditional means
- Simpson – continuous or binary predictor binary outcome
  - Difference in conditional and unconditional risk probability
- Simpson probably had nothing to do with it.
- Both date back to Pearson circa 1910
- Stigler's law

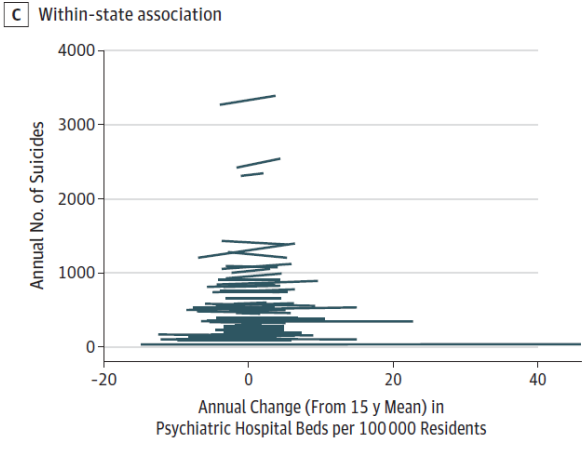
Making Sense of It

- Differences in results are due to adjustment.
- Since current weight (CW) is positively correlated with both BP and birth weight (BW), the relationship between BP and BW will change when CW is adjusted for.
- In RCTs covariate means are approximately equal – no between group difference (e.g., no correlation between BW and CW).

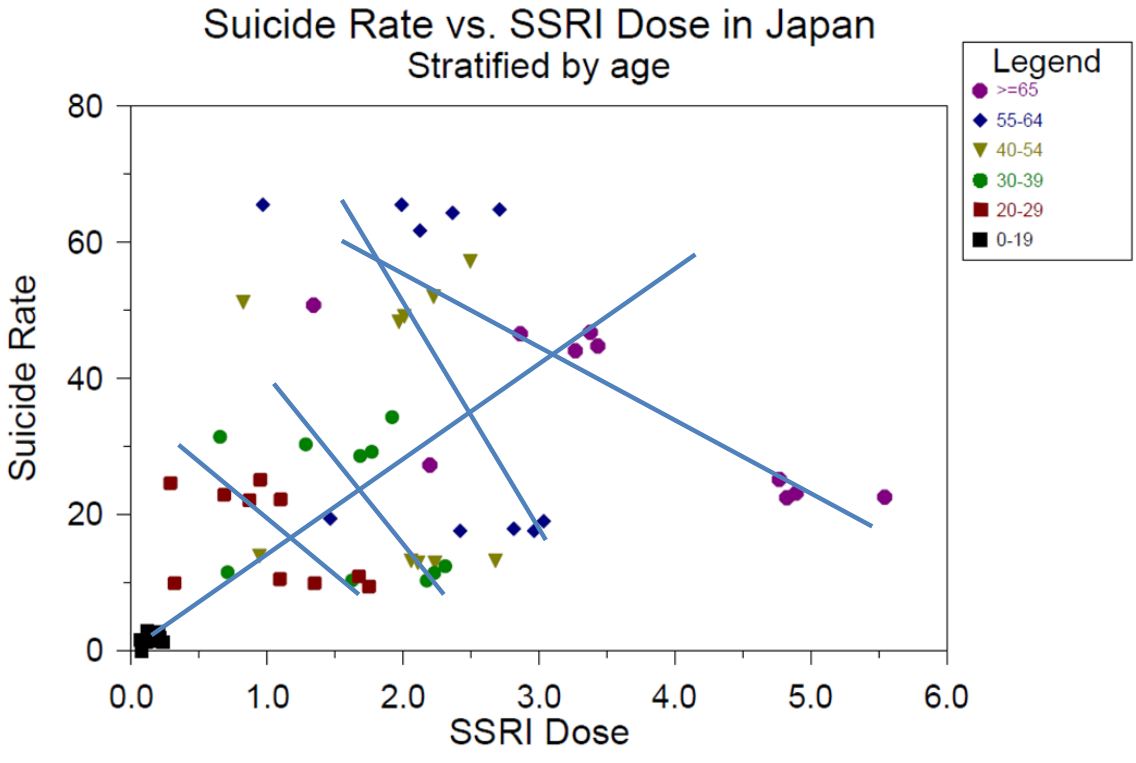
Suicide Rates and the Declining Psychiatric Hospital Bed Capacity in the United States







- Simpson's Paradox



Cochran (1968) *Biometrics* "Effectiveness of adjustment by subclassification in removing bias from observational studies"

Example - Age adjusted mortality rates vs smoking

	Nonsmokers	Cigarettes	Pipe and Cigar
Mortality per 1000 person-yrs %	13.5	13.5	17.4
2 subclasses	13.5	16.4	14.9
3 subclasses	13.5	17.7	14.2
10 subclasses	13.5	21.2	13.7

5-6 subclasses will remove 90% of bias.

The groups *must* overlap.

Nothing in output alerts you to nonoverlap.

70 year old smokers vs 40 year old nonsmokers.

Some Conceptual and Statistical Issues in Analysis of Longitudinal Psychiatric Data

Application to the NIMH Treatment of Depression Collaborative Research Program Dataset

Robert D. Gibbons, PhD; Donald Hedeker, PhD; Irene Elkin, PhD; Christine Waternaux, PhD; Helena C. Kraemer, PhD; Joel B. Greenhouse, PhD; M. Tracie Shea, PhD; Stanley D. Imber, PhD; Stuart M. Sotsky, MD; John T. Watkins, PhD

(Arch Gen Psychiatry. 1993;50:739-750) 711 citations

- Conceptual Foundations
  - Subject-specific effects
    - Heterogeneity
    - State Dependence
  - Missing data/dropouts
  - Patterns of correlation over time
  - Time-varying and invariant covariates

- Existing Methods
  - Endpoint analyses
  - ANOVA for Repeated Measures
  - MANOVA

TDCRP Study

- Treatment of Depression Collaborative Research Project
- Comparison of 2 types of psychotherapy
  - CBT vs IPT
- Added a pharmacotherapy control – imipramine
- Added placebo to control for drug
- 250 randomized, 239 entered, 219 post baseline
- Result based on LOCF – no difference between psychotherapy and pharmacotherapy

Dropouts

Table 1. Mean Baseline HRSD Scores for Sample Reaching Each Time Point*

	HRSD Score (N)				
	Week 0	Week 4	Week 8	Week 12	Week 16
CBT	19.6 (59)	19.7 (49)	19.4 (43)	19.1 (35)	19.2 (37)
IPT	19.6 (61)	19.2 (53)	18.8 (46)	18.9 (47)	18.9 (47)
IMI-CM	19.5 (57)	19.2 (48)	19.3 (44)	19.3 (39)	19.2 (37)
PLA-CM	19.5 (62)	19.0 (50)	19.0 (45)	19.1 (38)	19.1 (34)

**Rescreening scores. HRSD indicates Hamilton Rating Scale for Depression; CBT, cognitive behavior therapy; IPT, interpersonal psychotherapy; IMI-CM, imipramine with clinical management; and PLA-CM, placebo with clinical management.*

Observed Time Trends

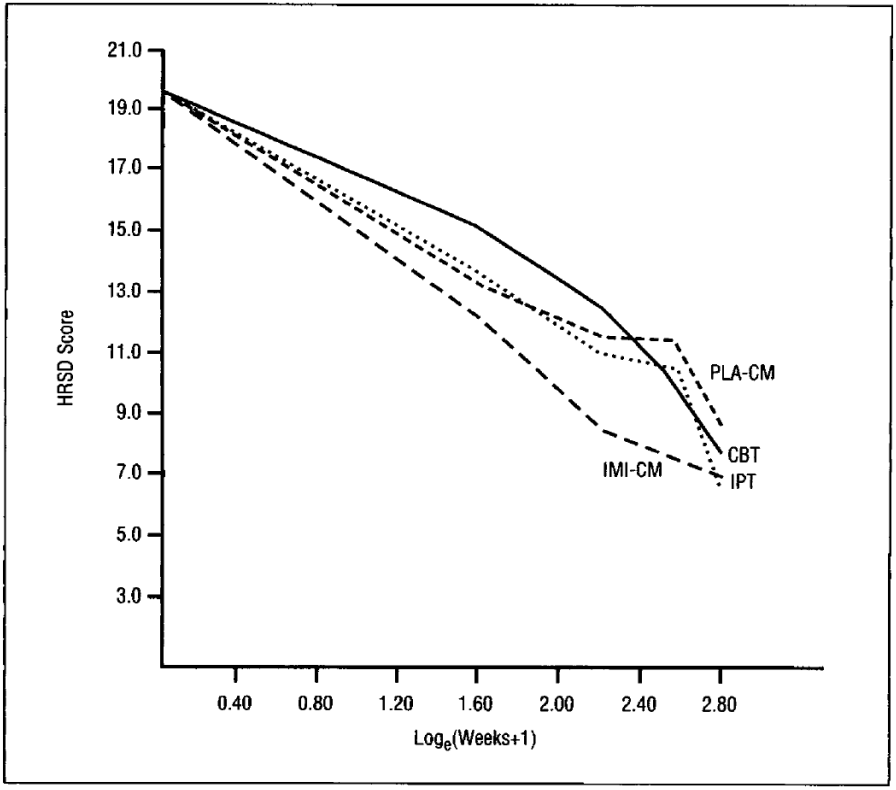


Figure 1. *Observed Hamilton Rating Scale for Depression (HRSD) treatment group means for available sample at each monthly measurement period. The x-axis represents time in $\log_e$(weeks+1); y-axis, average HRSD score for patients still in the study. CBT indicates cognitive behavior therapy; IPT, interpersonal psychotherapy; IMI-CM, imipramine with clinical management; and PLA-CM, placebo with clinical management. The measured time points: weeks 4, 8, 12, and 16 correspond to x-axis values of 1.61, 2.20, 2,57, and 2.83, respectively, in the scale of $\log_e$(weeks+1).*

Estimated Time Trends

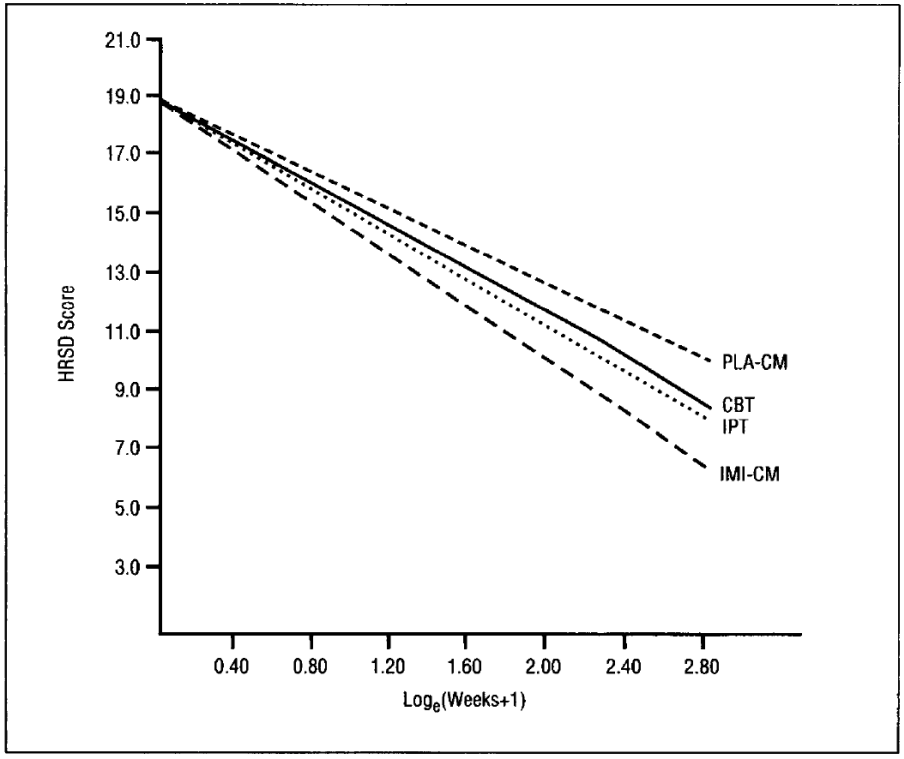


Figure 2. *Estimated Hamilton Rating Scale for Depression (HRSD) trend lines by treatment group using all available data for each patient. The x-axis represents time in $\log_e$(weeks+1); y-axis, average HRSD score. CBT indicates cognitive behavior therapy; IPT, interpersonal psychotherapy; IMI-CM, imipramine with clinical management; and PLA-CM, placebo with clinical management. The measured time points: weeks 4, 8, 12, and 16 correspond to x-axis values of 1.61, 2.20, 2,57, and 2.83, respectively, in the scale of $\log_e$(weeks+1).*

Polls 10 and 11

Missing Data

Table 4. Empirical Bayes Estimates of Missing Data*

Patient	Group	Slope	Week				
			0	4	8	12	16
32	PLA	−1.15	17	23	17	20	**17.3**
57	PLA	−0.40	27	26	26	23	**19.8**
172	CBT	−0.72	24	30	24	**19.9**	**17.5**
142	PLA	−2.70	14	16	**14.2**	**13.1**	**12.4**
218	IMI	−2.97	20	15	17	15	**12.1**
160	IMI	−4.01	22	20	**13.0**	5	9
211	PLA	−2.13	21	**16.5**	14	19	**15.1**
133	IMI	−5.66	20	10	7	2	**2.7**
206	IMI	−5.97	22	9	3	**2.8**	1
233	PLA	−5.43	20	11	4	**4.1**	0
114	CBT	−4.79	21	8	8	**7.3**	**6.7**
226	IPT	−4.87	25	10	**8.2**	**6.9**	**5.6**
161	IMI	−5.91	22	5	**4.5**	**4.1**	**2.5**

**PLA indicates placebo; CBT, cognitive behavior therapy; IMI, imipramine; and IPT, interpersonal psychotherapy. Entries in normal typeface are observed values; entries in boldface type, model estimates of missing data.*